



Stored Procedures(SP)



What is Stored Procedures

A stored procedure is nothing more than prepared SQL code that you save so you can reuse the code over and over again. So if you think about a query that you write over and over again, instead of having to write that query each time you would save it as a stored procedure and then just call the stored procedure to execute the SQL code that you saved as part of the stored procedure.



Why to use stored procedures and advantages of stored procedures.

A stored procedure is a group of Transact-SQL statements compiled into a single execution plan.

Stored Procedures are coding block in database server. It is pre compiled entity i.e. it is compiled at once and can be used again and again.

With help of stored procedure a group of SQL statements can be executed sequentially.

To supply data to the procedure we must have to use parameters in procedure.

Stored procedures use parameters mapping concept.

In parameter mapping front end and procedure parameters names, type and direction must be same and where front-end parameter length should be less than or equal to procedure parameter length (than only can map parameters).

To return any value from procedure we use return statement.



Types Of Stored Procedures

User Defined Stored procedure: The user defined stored procedures are created by users and stored in the current database

system Stored Procedure: The system stored procedure have names prefixed with sp_. Its manage SQL Server through administrative tasks. Which databases store system stored procedures are master and msdb databases.

Parameter in Stored Procedures



Without Parameter:

```
Create procedure WitoutparameterExp
As
Begin

Select ed.EmpID, e.EmpName,ed.DeptID,e.EmpSalary
from EmployeeDetails ed
Inner Join EmpSalary e on e.EmpID = ed.EmpID

Select ed.EmpID,ed.EmpName,EmpSalary,ed.DeptID,d.DepartmentName
from EmployeeDetails ed
inner Join EmpSalary e on e.EmpID = ed.EmpID
Inner Join Department d on d.DeptID = ed.DeptID

End
go

--

exec WitoutparameterExp
```

With Parameter:

```
Create procedure parameterExp(@EmpID int)
As
Begin

Select ed.EmpID, e.EmpName,ed.DeptID,e.EmpSalary
from EmployeeDetails ed
Inner Join EmpSalary e on e.EmpID = ed.EmpID
Where ed.EmpID = @EmpID

Select ed.EmpID,ed.EmpName,EmpSalary,ed.DeptID,d.DepartmentName
from EmployeeDetails ed
inner Join EmpSalary e on e.EmpID = ed.EmpID
Inner Join Department d on d.DeptID = ed.DeptID
Where ed.EmpID = @EmpID

End
go
```

```
-----Two Output Parameter-----
create PROCEDURE GetstudentnameInOutputVariable
(
    @studentid INT,                --Input parameter , Studentid of the student
    @studentname VARCHAR (200) OUT, -- Output parameter to collect the student name
    @StudentEmail VARCHAR (200)OUT  -- Output Parameter to collect the student email
)
AS
BEGIN
```